



SUN YAT-SEN UNIVERSITY

Presentation for PhD Application

Applicant: Zihao Jin

School of Artificial Intelligence · Sun Yat-sen University





My Background

Zihao Jin 2005.1 Shenzhen, China

Educational Background

Sun Yat-sen University Shenzhen, China

- Bachelor of Engineering in Artificial Intelligence (supervisor: Ruimao Zhang)
- GPA: 86/100 09/2023--present

Core Courses

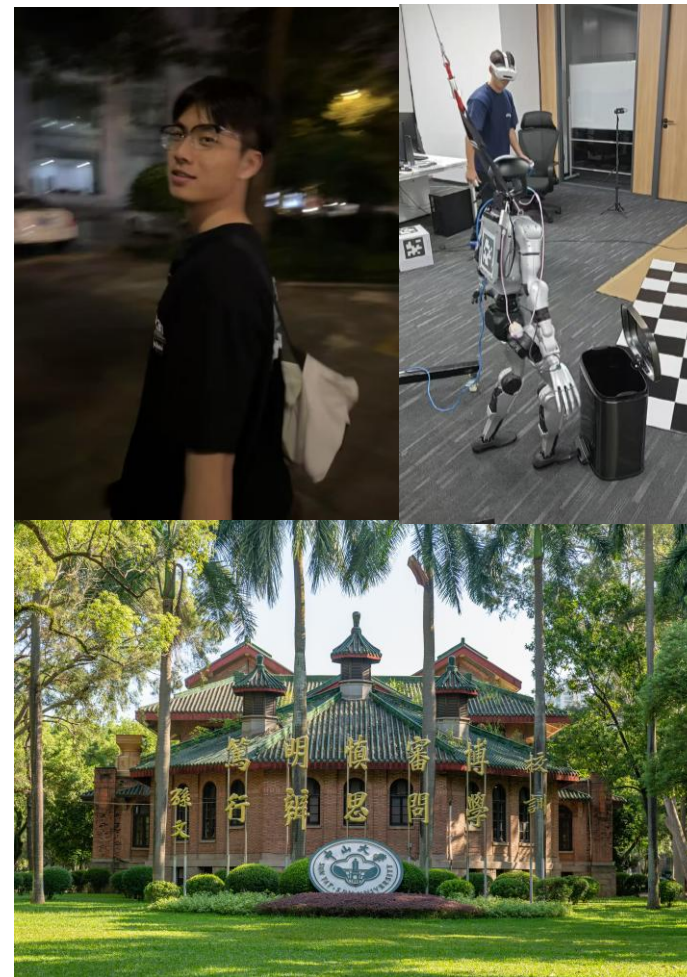
Computer Vision (97), Machine Learning (96), Human-Computer Interaction (94), Numerical Computing (96), Digital Signal Processing (94), Game Theory (90)

Awards

- SYSU Outstanding Student Scholarship
- Provincial Mathematical Modeling Award
- National Collegiate Dragon Boat Championship

Publications

Nereus: A Large-Scale Underwater Dataset for Fine-Grained Attribute Understanding and Grounded Counting
Research Project, NeurIPS2026 Under review





My Research Journey: From Perception to Adaptive Skills

Multimodal Perception → Embodied Robot Learning → Adaptive Visuomotor Intelligence

COMPLETED

Nereus

Grounded multimodal perception

01

Build and evaluate structured underwater visual-language supervision.

ONGOING

Ego2Robot

Real-world robot experience

02

Collect synchronized demonstrations for humanoid policy learning.

PROPOSED

EvoSkill-Agent

Lifelong skill evolution

03

Turn experience and failure into reusable, improving robot skills.

One trajectory: perceive with evidence, act through experience, improve through reusable skills.



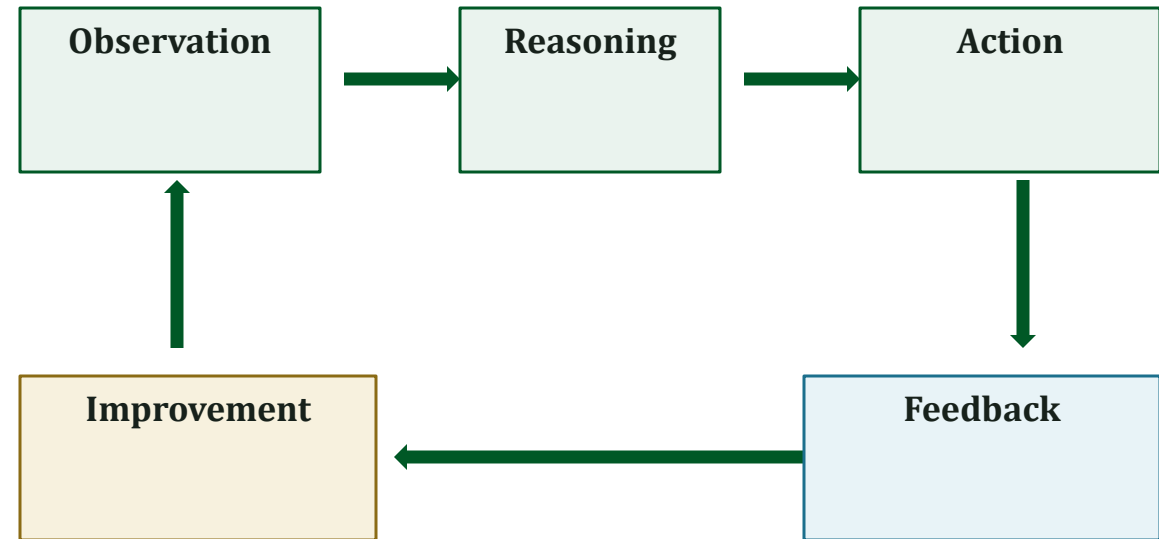
Robot Intelligence Requires Closing the Perception-Action Loop

ONE-WAY PREDICTION



Perception explains what is present and grounds a prediction in visual evidence.

CLOSED-LOOP EMBODIED INTELLIGENCE

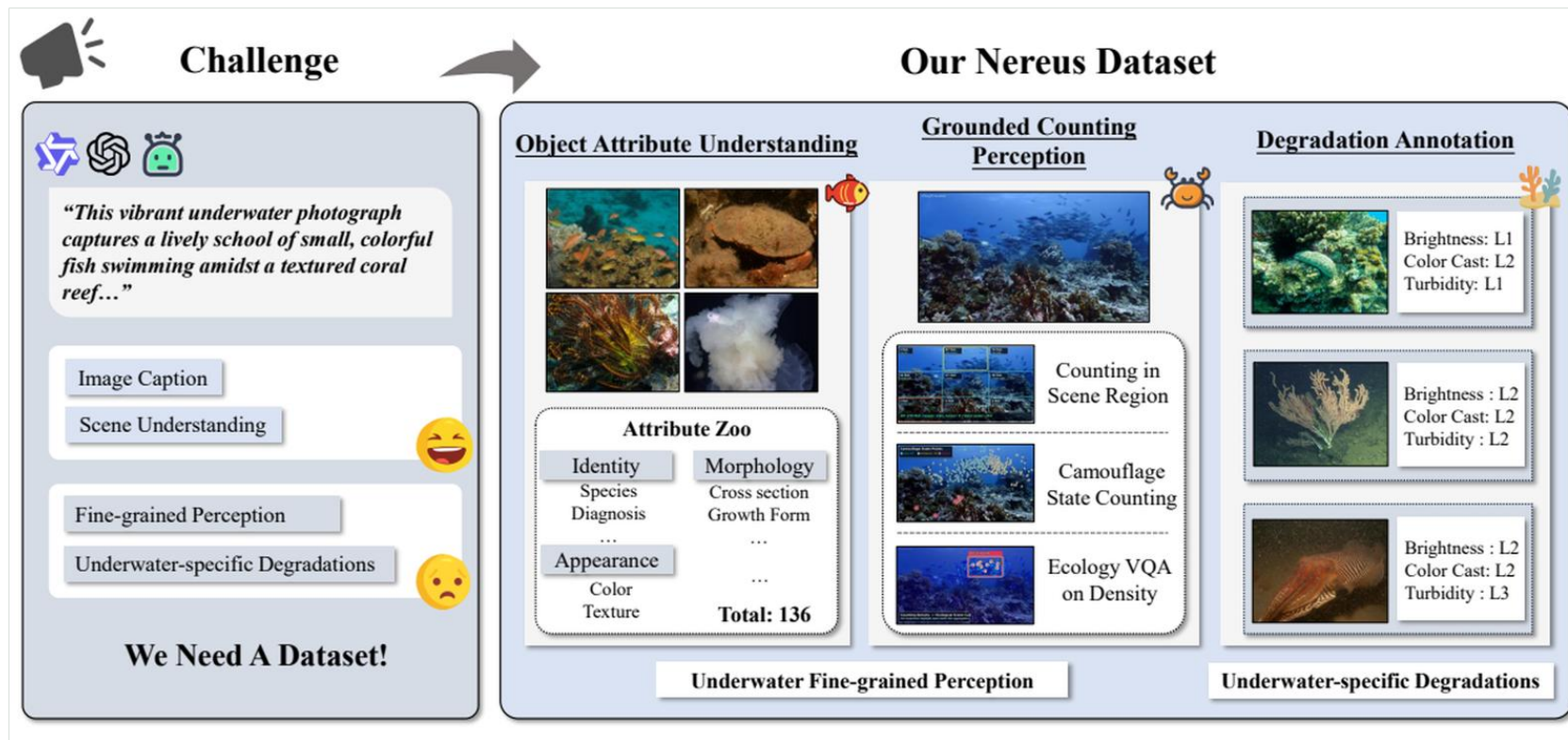


Embodied intelligence chooses an action, observes the outcome, and uses feedback to improve.

My research focuses on bridging perception and action for adaptive robots.

Nereus Turns Underwater Images into Grounded Evidence

COMPLETED RESEARCH



LIMITATION

Existing underwater resources often emphasize captions, coarse categories, or total counts.

NEREUS CONTRIBUTION

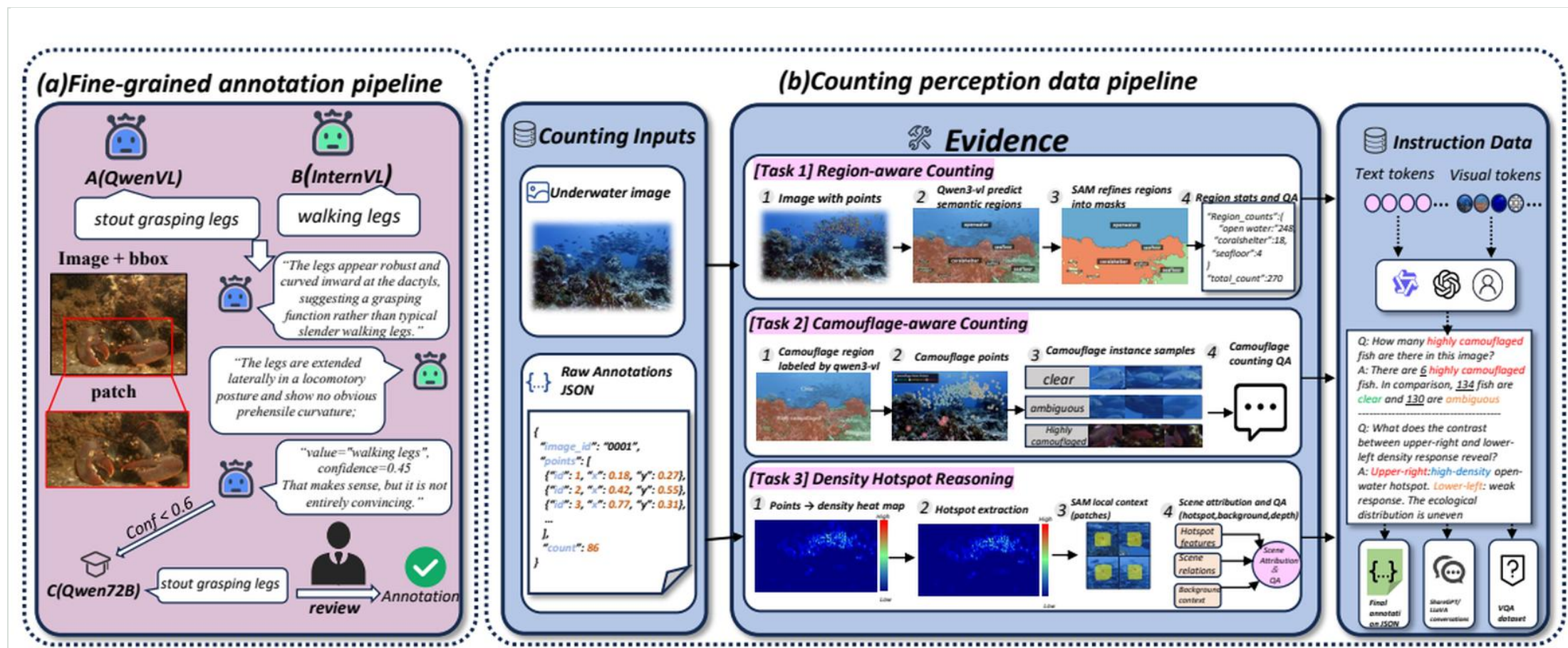
- Fine-grained object attributes
- Grounded counting linked to image evidence
- Degradation-aware interpretation

Structured supervision connects an answer to what is present, where it is, and how image quality affects interpretation.

Building Nereus Required an End-to-End Research Pipeline

COMPLETED RESEARCH

QUALITY CONTROL · Human-calibrated annotation review



88K

images

2.68M

QA pairs

136K

objects

2,690

species

65K

point annotations

The Next Question Is Not Only What Is It? but What Should the Robot Do?

TRANSITION TO ONGOING EMBODIED WORK

GROUNDING PERCEPTION

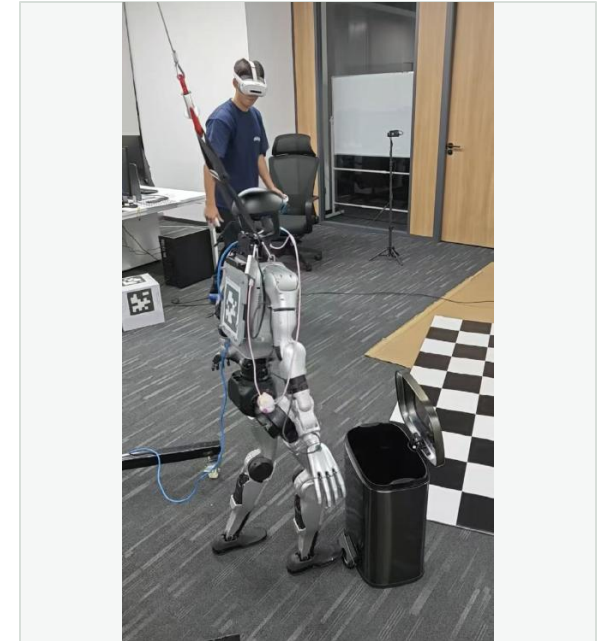


What is present, and what evidence supports the answer?

Perception + Physical interaction + Learning

From evidence to embodied intelligence

PHYSICAL INTERACTION

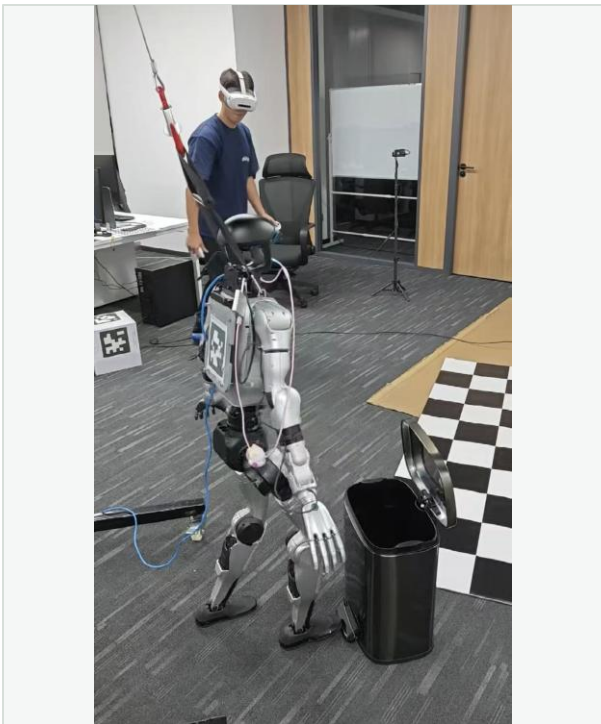


What action should the robot execute, and how should feedback change the next action?

Grounded perception becomes more useful when it supports action and adaptation in the physical world.

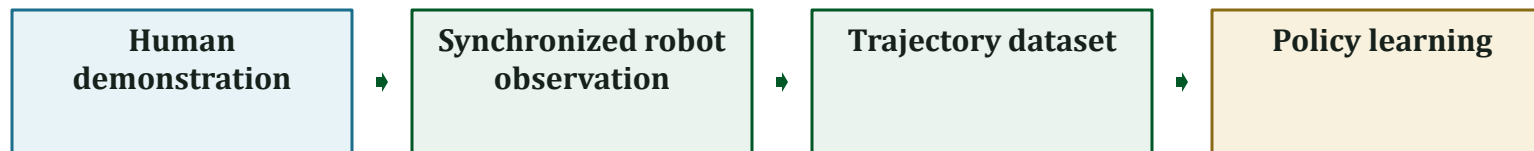
Ego2Robot Establishes a Real-World Humanoid Learning Pipeline

ONGOING EMBODIED WORK



Authentic PICO teleoperation photograph with Unitree G1

EDITABLE DATA PIPELINE



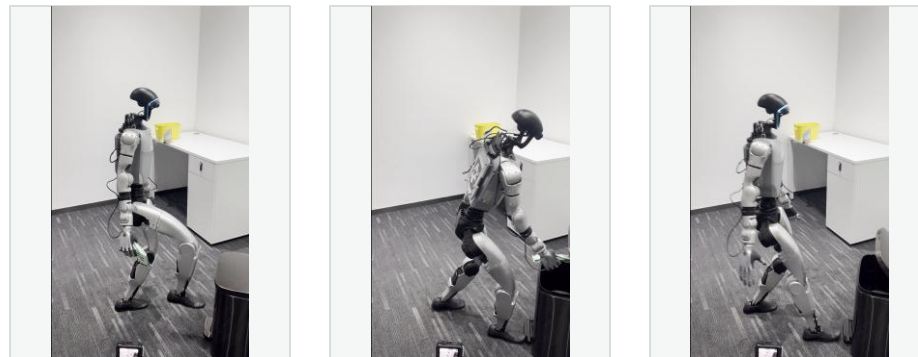
HARDWARE / INTERFACE

Unitree G1

RealSense D435i

PICO teleoperation

DEPLOYMENT EVIDENCE



1 s

3 s

5 s

The current evidence establishes teleoperation, synchronized observation, and trajectory collection; downstream policy results remain in progress.



Current Policies Learn Actions, but Rarely Accumulate Skills

PROPOSED RESEARCH GAP

ONE-PASS POLICY PIPELINE



A policy produces an action, then the pipeline moves on.

EXPERIENCE → MEMORY → IMPROVEMENT



Execution outcomes return to reusable skill memory.

WHAT THE GAP MOTIVATES

- Reusable skill abstraction from multimodal experience
- Skill retrieval and composition for long-horizon tasks
- Learning from execution failure during deployment

The gap concerns practical deployment pipelines that do not explicitly retain and improve reusable skills.

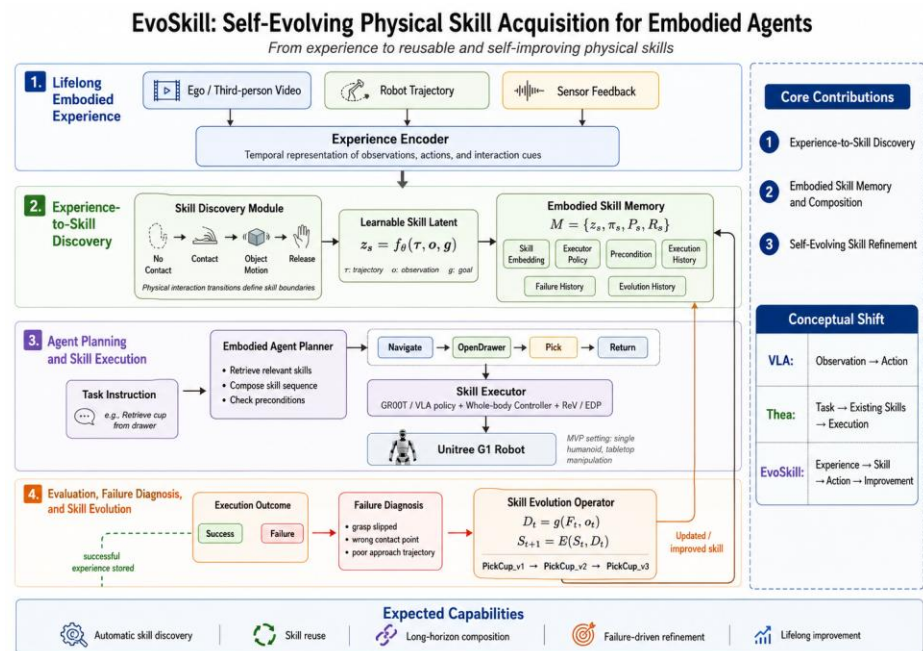
Future Plan

Research Interest

- Embodied Skill Learning for Adaptive Humanoid Robots
- Adaptive Visuomotor Intelligence for Embodied Robots

My goal is to develop robots that understand the physical world, acquire reusable skills, and improve through interaction.

Proposed Research Direction



Skill Discovery

How can robots extract reusable physical skills from multimodal experience?

OBSERVABLE OUTPUT

Skill reuse

Skill Memory

How can robots retrieve and compose skills for long-horizon tasks?

OBSERVABLE OUTPUT

Composition success

Skill Evolution

How can execution feedback diagnose failure and improve future behavior?

OBSERVABLE OUTPUT

Failure-recovery improvement

The core of **EvoSkill** is to enable a robot to convert accumulated embodied experience into reusable physical skills, and then continuously improve those skills through execution feedback.



Skills

Skills

- Languages & Tools: Python, C++, MATLAB, Linux, Git, Gradio
- ML / Vision: PyTorch, OpenCV, multimodal LLMs/VLMs, Diffusion Policy/EDP, object detection, pose estimation
- Robotics / Data: Isaac Lab, GR00T, LeRobot, RealSense, ZMQ, data collection, benchmark construction, model evaluation
- Languages: Mandarin Chinese (native); English: CET-4 / CET-6.

Self Assessment



Well-motivated



Analytical



Diligent



Inquisitive



Thanks!

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